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Jawaban

$$y = f(x) = x^3 - 2x + 3$$

$$\Delta y = f(x + \Delta x) - f(x)$$

dengan  $x = 2$ ,  $\Delta x = 0.1$ 

$$\Rightarrow \Delta y = f(2.1) - f(2)$$

$$= ((2.1)^3 - 2(2.1) + 3) - (2^3 - 2(2) + 3)$$

$$= 8.061 - 7$$

$$= \underline{1.061}$$

$$\frac{dy}{dx} = 3x^2 - 2$$

$$\Rightarrow dy = (3x^2 - 2)dx$$

$$= (3(2)^2 - 2)(0.1)$$

$$= 10.01$$

$$= \underline{1}$$

$$\text{Galat (rel)} = \left| \frac{dy - \Delta y}{\Delta y} \right| = \left| \frac{1 - 1.061}{1.061} \right|$$

$$= 5.7492\%$$

$$r = 12 \text{ cm}, \Delta r = 0.5 \text{ cm} \Rightarrow \bar{r} = (12 \pm 0.5) \text{ cm}$$

$$V = \frac{4}{3}\pi r^3 \Rightarrow \Delta V = 4\pi r^2 \Delta r$$

$$= 4\pi (12)^2 \cdot 0.5$$

$$= \underline{288\pi \text{ cm}^3}$$

\* Pakai notasi fisika,  
yaitu  $\Delta$ , bukan  $d$

$$\text{Galat relatif} = \frac{\Delta V}{V} = \frac{288\pi}{\frac{4}{3}\pi (12)^3} \times 100\%$$

$$= \frac{288\pi}{2304\pi} \times 100\%$$

$$= \frac{1}{8} \times 100\%$$

$$= \underline{12.5\%}$$

3. Mencari nilai maksimum dan minimum

(a)  $f(x) = 2x^3 - 3x^2 - 12x + 1$   $[-2, 3]$

$$f(-2) = 2(-2)^3 - 3(-2)^2 - 12(-2) + 1$$

$$= -3$$

$$f(3) = 2(3)^3 - 3(3)^2 - 12(3) + 1$$

$$= -8$$

$$f'(x) = 6x^2 - 6x - 12 = 0$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x-2)(x+1) = 0 \Rightarrow x \in \{-1, 2\}$$

$$\hookrightarrow f(2) = 2(2)^3 - 3(2)^2 - 12(2) + 1$$

$$= -19$$

$$\hookrightarrow f(-1) = 2(-1)^3 - 3(-1)^2 - 12(-1) + 1$$

$$= 8$$

maksimum =  $\boxed{8}$  saat  $x = -1$

minimum =  $\boxed{-19}$  saat  $x = 2$

(b)  $f(x) = \frac{x}{x^2 - x + 1}$   $[0, 3]$

$$f(0) = 0, f(3) = \frac{3}{7}$$

Bukti bahwa  $x^2 - x + 1 \neq 0$  untuk  $[0, 3]$

$$\frac{d}{dx} x^2 - x + 1 \Rightarrow 2x - 1, \downarrow [0, \frac{1}{2}), \uparrow [\frac{1}{2}, 3]$$

Saat  $x=0$ ,  $x^2 - x + 1 = 1$

$x = \frac{1}{2}$ ,  $x^2 - x + 1 = \frac{1}{2}$ , menurut Teorema Nilai Antara,  
tidak ada nilai 0 untuk  
 $x \in [0, \frac{1}{2})$

karena  $x^2 - x + 1$  meningkat untuk  $x \in [\frac{1}{2}, 3]$ ,  $\Rightarrow$

$x^2 - x + 1 > 0$  sehingga tidak ada asimtot tegak  
untuk  $f(x)$  pada  $[0, 3]$

$$f'(x) = 0 \Rightarrow (x^2 - x + 1) - x(2x - 1) = 0 \Rightarrow x = 1$$

maksimum =  $\boxed{1}$  saat  $x = 1$

minimum =  $\boxed{0}$  saat  $x = 0$

$$(c) f(t) = 2 \cos t + \sin(2t) \quad [0, 2]$$

$$f(0) = 2, f'(t) = 0$$

$$f'(t) = -2 \sin t + 2 \cos 2t = 0$$

$$\Rightarrow \cos 2t - \sin t = 0$$

$$\Rightarrow (1 - 2 \sin^2 t) - \sin t = 0$$

$$\Rightarrow -2 \sin^2 t - \sin t + 1 = 0$$

$$\Rightarrow 2 \sin^2 t + \sin t + 1 = 0$$

$$\Rightarrow (2 \sin t - 1)(\sin t + 1) = 0$$

$$\text{Pada } [0, \frac{\pi}{2}], \text{ maka } \sin t = \frac{1}{2} \Rightarrow t = \frac{\pi}{6}$$

$$f(\frac{\pi}{6}) = \frac{3\sqrt{3}}{2}$$

$$27 > 16 \Leftrightarrow \sqrt{27} > 4$$

$$\Leftrightarrow 3\sqrt{3} > 4$$

$$\Leftrightarrow \frac{3\sqrt{3}}{2} > 2$$

$$\text{maksimum} = \left[ \frac{3\sqrt{3}}{2} \right] \text{ saat } t = \frac{\pi}{6}$$

$$\text{minimum} = [0] \text{ saat } t = \frac{\pi}{2}$$

4. Mencari selang kemonotonan, kecekungan, dan titik infleksi

$$(a) f(x) = x^4 - 2x^2 + 3$$

$$f'(x) = 4x^3 - 4x = 0$$

$$\Rightarrow 4x(x^2 - 1) = 0 \Rightarrow x \in \{-1, 0, 1\} \rightarrow (-\infty, -1), (-1, 0), (0, 1), (1, \infty)$$

$$(-\infty, -1) \text{ cek } x = -2 \Rightarrow f'(-2) = -24 < 0 \Rightarrow \text{turun}$$

$$(-1, 0) \text{ cek } x = -\frac{1}{2} \Rightarrow f'(-\frac{1}{2}) = 1,5 > 0 \Rightarrow \text{naik}$$

$$(0, 1) \text{ cek } x = \frac{1}{2} \Rightarrow f'(\frac{1}{2}) = -1,5 < 0 \Rightarrow \text{turun}$$

$$(1, \infty) \text{ cek } x = 2 \Rightarrow f'(2) = 24 > 0 \Rightarrow \text{naik}$$

Maka, selang

$(-\infty, -1)$  monoton turun

$(-1, 0)$  monoton naik

$(0, 1)$  monoton turun

$(1, \infty)$  monoton naik

$$\text{monoton naik} = (-1, 0) \cup (1, \infty)$$

$$\text{monoton turun} = (-\infty, -1) \cup (0, 1)$$

$$f''(x) = 12x^2 - 4 = 0 \Rightarrow x \in \{-\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}\}$$

$(-\infty, -\frac{1}{3}\sqrt{3})$  coba  $x = -1 \Rightarrow f''(-1) = 8 > 0 \Rightarrow$  cekung ke atas

$(-\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3})$  coba  $x = 0 \Rightarrow f''(0) = -4 < 0 \Rightarrow$  cekung ke bawah

$(\frac{1}{3}\sqrt{3}, \infty)$  coba  $x = 1 \Rightarrow f''(1) = 8 > 0 \Rightarrow$  cekung ke atas

Maka, selang

|                                                               |                                                                               |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|
| $(-\infty, -\frac{1}{3}\sqrt{3})$ cekung ke atas              | cekung = $(-\infty, -\frac{1}{3}\sqrt{3}) \cup (\frac{1}{3}\sqrt{3}, \infty)$ |
| $(-\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3})$ cekung ke bawah |                                                                               |
| $(\frac{1}{3}\sqrt{3}, \infty)$ cekung ke atas                | cekung = $(-\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3})$                        |

dengan titik infleksi

$$(-\frac{1}{3}\sqrt{3}, \frac{22}{9}) \text{ dan } (\frac{1}{3}\sqrt{3}, \frac{22}{9})$$

(b)  $f(x) = \frac{x}{x^2+1}$

$$f'(x) = \frac{(x^2+1) - x(2x)}{(x^2+1)^2} = 0$$

$$\Rightarrow x^2 = 1 \Rightarrow x \in \{-1, 1\} \rightarrow (-\infty, -1), (-1, 1), (1, \infty)$$

$$f'(-2) = \frac{\text{neg}}{\text{pos}} \Rightarrow \text{turun}$$

$$f'(0) = \frac{1}{\text{pos}} \Rightarrow \text{naik}$$

$$f'(2) = \frac{\text{neg}}{\text{pos}} \Rightarrow \text{turun}$$

Maka, selang

|                               |                                                              |
|-------------------------------|--------------------------------------------------------------|
| $(-\infty, -1)$ monoton turun | turun = $(-\infty, -1) \cup (1, \infty)$<br>naik = $(-1, 1)$ |
| $(-1, 1)$ monoton naik        |                                                              |
| $(1, \infty)$ monoton turun   |                                                              |

$$f''(x) = \frac{-2x(x^2+1)^2 - (1-x^2) \cdot 2(2x)(x^2+1)}{(x^2+1)^4} = 0$$

$$\Rightarrow x \in \{-\sqrt{3}, 0, \sqrt{3}\} \rightarrow (-\infty, -\sqrt{3}), (-\sqrt{3}, 0), (0, \sqrt{3}), (\sqrt{3}, \infty)$$

$$f''(-2) = -4 < 0 \Rightarrow \text{cekung ke bawah}$$

$$f''(-1) = \dots \text{atas}$$

[Sudah pasti selang-selang] bawah  
karena garis ganjil atas

Maka, selang

|                                        |                                                    |
|----------------------------------------|----------------------------------------------------|
| $(-\infty, -\sqrt{3})$ cekung ke bawah | cekung = $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$ |
| $(-\sqrt{3}, 0)$ cekung ke atas        |                                                    |
| $(0, \sqrt{3})$ cekung ke bawah        | cekung = $(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty)$  |
| $(\sqrt{3}, \infty)$ cekung ke atas    |                                                    |

semua  
atau ganjil

dengan titik-titik infleksinya.

$$\left(-\sqrt{3}, -\frac{1}{4}\sqrt{3}\right), (0, 0), \text{ dan } \left(\sqrt{3}, \frac{1}{4}\sqrt{3}\right)$$

(c)  $f(x) = \cos^2 x - 2 \sin x \quad [0, 2\pi]$

$$\sin^2 x + \cos^2 x = 1 \Rightarrow \cos^2 x = 1 - \sin^2 x$$

$$\Rightarrow f(x) = 1 - \sin^2 x - 2 \sin x$$

$$f'(x) = -\sin(2x) - 2 \cos x = 0$$

$$\Rightarrow -2 \cos x (\sin x + 1) = 0$$

$$\Rightarrow x \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\} (\cos x = 0)$$

$$\in \left\{\frac{3\pi}{2}\right\} (\sin x = -1)$$

$$\Rightarrow x \in \cup_{\cos x = 0 \text{ dan } \sin x = -1} \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\} \rightarrow \left(-\infty, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), \left(\frac{3\pi}{2}, \infty\right)$$

$$f'(0) = -2 < 0 \text{ turun}$$

|       |                                                           |              |
|-------|-----------------------------------------------------------|--------------|
| Maka, | Selang $\left[0, \frac{\pi}{2}\right)$ monoton turun      | } akar ganda |
|       | $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ monoton naik |              |
|       | $\left(\frac{3\pi}{2}, 2\pi\right]$ monoton turun         |              |

$$f''(x) = -2 \cos(2x) + 2 \sin x = 0$$

$$\Rightarrow \sin x - \cos 2x$$

$$= \sin x - (1 - 2 \sin^2 x)$$

$$\in (2 \sin x - 1)(\sin x + 1) = 0$$

$$\Rightarrow \sin x = \frac{1}{2} \text{ atau } \sin x = -1$$

$$\Rightarrow x \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}, x \in \left\{\frac{3\pi}{2}\right\}$$

$$\Rightarrow x \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}$$

$f''(0) : \text{neg} \Rightarrow \text{cekung ke bawah}$

|       |                                                               |                                                                                          |
|-------|---------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Maka, | Selang $\left[0, \frac{\pi}{6}\right)$ cekung ke bawah        | } $\text{cekung} = \left[0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right]$ |
|       | $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ cekung ke atas   |                                                                                          |
|       | $\left(\frac{5\pi}{6}, \frac{3\pi}{2}\right)$ cekung ke bawah | } akar ganda                                                                             |
|       | $\left(\frac{3\pi}{2}, 2\pi\right]$ cekung ke bawah           |                                                                                          |

dengan titik infleksi

$$\left(\frac{\pi}{6}, -\frac{1}{4}\right) \text{ dan } \left(\frac{5\pi}{6}, -\frac{1}{4}\right)$$

5. Nilai-nilai maksimum dan minimum lokal

(a)  $f(x) = -2x^2 + 3x^2 + 1$

$$f'(x) = -6x^2 + 6x = 0$$

$$\Rightarrow x(x-1) = 0 \Rightarrow x \in \{0, 1\}$$

$$f''(x) = -12x + 6 \Rightarrow f''(0) = 6 > 0, \text{ minimum lokal} = f(0)$$

$$f''(1) = -6 < 0, \text{ maximum lokal} = f(1)$$

$$\text{Minimum lokal} = 1 \text{ saat } x=0$$

$$\text{Maximum lokal} = 2 \text{ saat } x=1$$

(b)  $f(x) = \frac{x^2}{x-1}$

$$f'(x) = \frac{2x(x-1) - x^2}{(x-1)^2} = 0$$

$$\Rightarrow x(2(x-1) - x) = 0$$

$$\Rightarrow x(x-2) = 0 \Rightarrow x \in \{0, 2\}$$

$$f''(x) = \frac{(2x-2)(x-1)^2 - 2(x-1)(x^2-2x)}{(x-1)^4}$$

$$f''(0) = -2 \cdot 1 = -2 < 0, \text{ maksimum} \Rightarrow f(0) \text{ maks}$$

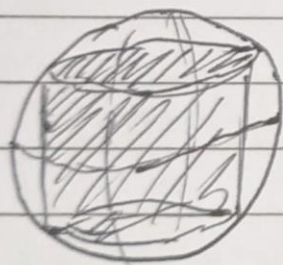
$$f''(2) = 2 > 0, \text{ minimum} \Rightarrow f(2) \text{ min}$$

$$\text{minimum lokal} = 4 \text{ saat } x=2$$

$$\text{maksimum lokal} = 0 \text{ saat } x=0$$

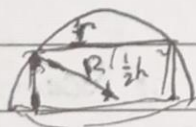
maks lokal tidak harus lebih besar daripada min lokal

6.



$$V_b = \frac{4}{3}\pi R^3, R=10$$

$$V_t = \pi r^2 h, \text{ maks } V_t?$$



Pythagoras

$$\Rightarrow r^2 + \left(\frac{1}{2}h\right)^2 = R^2$$

$$\Rightarrow r^2 + \frac{h^2}{4} = 100$$

$$\Rightarrow r^2 = 100 - \frac{h^2}{4} \quad (1)$$

$$V_t = \pi r^2 h$$

$$= \pi \left(100 - \frac{h^2}{4}\right) \cdot h$$

$$= 100\pi h - \frac{\pi h^3}{4}$$

$$V'_t = 0 \Rightarrow h = \frac{20}{3}\sqrt{3}, \text{ pakai persamaan (1) didapat}$$

$$r = \frac{10}{3}\sqrt{6}$$

Dengan demikian

$$h = 20\sqrt{3}$$

3

$$r = 10\sqrt{6}$$

3

dan volume maksimumnya

$$V_{\max} = \frac{4000\pi\sqrt{3}}{9}$$

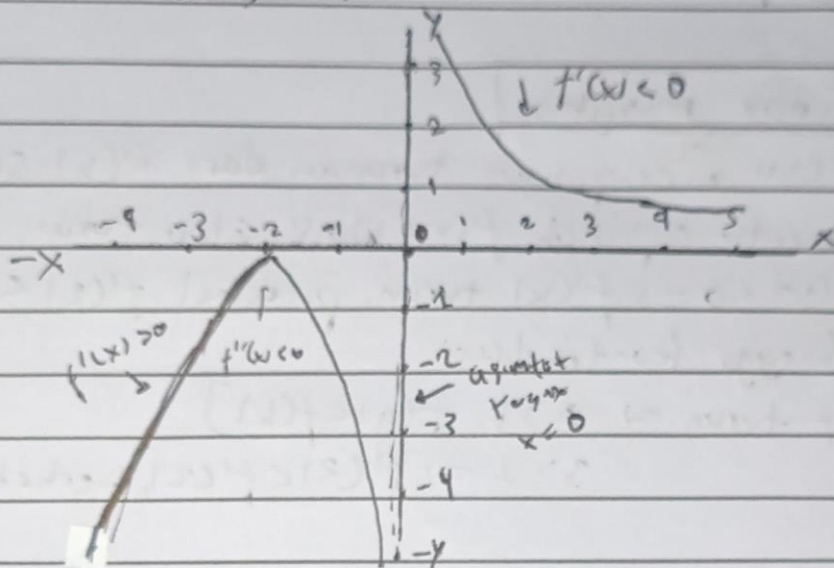
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7. Kurva fungsi cisarut tegak  $x=0$ , naik  $(-\infty, -1)$

cekung bawah  $x < 0$  turun  $(-2, 0) \cup (0, \infty)$

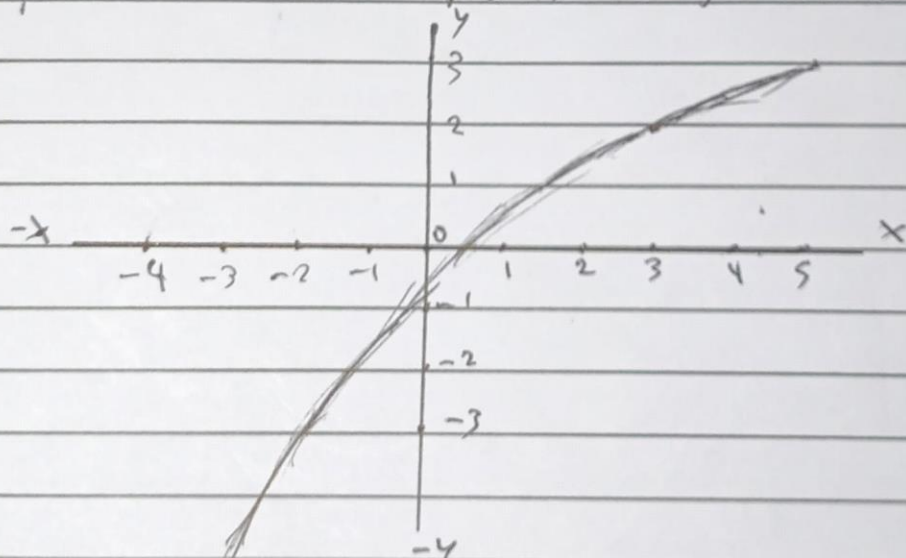
cekung atas  $x > 0$



8.  $f(3) = 2$ ,  $f'(3) = \frac{1}{2}$ ,  $f'(x) > 0$ ,  $f''(x) < 0$

nilai  $f(3) = 2$ , naik saat  $f(3)$ , cekung ke bawah  $\forall x$

(a)



(b)  $f(x) = 0$  hanya memiliki satu solusi

Karena  $\forall x, f'(x) > 0$  fungsi  $\tilde{x}$  tidak mungkin menyentuh garis  $y=0$  dua kali sehingga solusinya  $< 2$

Anggap  $f(x) \neq 0$  (no solusi), maka akan ada asimtot datar pada  $f(x)$  padahal kurva cekung ke bawah sehingga pada  $x \rightarrow -\infty$  akan terturun ke  $y \rightarrow -\infty$ .

Berdasarkan Teorema Nilai Antara, karena  $f$  terturunkan  $\Rightarrow$  kontinu ada  $c$  sehingga  $f(c) = 0$ .

■

(c) Tidak mungkin

$f''(x)$  merupakan turunan dari  $f'(x)$  sehingga merupakan penentu apakah  $f'(x)$  naik atau turun. Diketahui bahwa  $f''(x) < 0 \Rightarrow f'(x)$  turun, padahal  $f'(2) < f'(3)$

Sehingga kontradiksi

[ $f$  turun  $\Rightarrow a > b, f(a) < f(b)$ ]

$3 > 2 \Rightarrow f'(3) < f'(2)$ , berbeda dengan  $\frac{1}{3} < \frac{1}{2}$

■